

Physical parameters of the G4VM application

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1 Recorded quantities: /sd/format reference

1.1 The three output levels

/sd/verbose is a bit mask, and each bit writes a different file with a different degree of aggregation:

Bit	File	One row per
4	*.steps	Geant4 step inside a sensitive volume
2	*.tracks	(detector, track) pair
1	*.hits	detector, summed over all its tracks

Aggregation is not uniform across keys, and this is the single most common source of misinterpretation. Only five quantities are additive and are actually summed: **ioni**, **nioni**, **len**, **optPhot**, **scintPhot**. Two are deliberately overwritten with the value of the *last* step of the track: **post_dir** and **post_proc**. **Every remaining key keeps the value it had on the *first* step of that track in that detector.** At hit level the surviving non-summed fields are those of the track with the *lowest track ID* that entered the detector — usually, but not necessarily, the primary.

Legend for the three rightmost columns of the reference table:

Symbol	Meaning at that level
=	invariant over what was aggregated — the value is exact
Σ	summed over the aggregated steps (track) or tracks (hit)
◦	frozen: value of the <i>first</i> step (hit: of the lowest-ID track)
*	updated: value of the <i>last</i> step (hit: of the lowest-ID track)

1.2 Reference table

Units are those actually written to the file. Keys marked ×3 expand to three whitespace-separated columns, which must be accounted for when parsing. Keys marked in **red** are output from the custom implementation of the cherenkov light production and scintillation light yield further described in Chapters 2 and 3.

Table 1: Complete list of /sd/format keys.

Key	Unit	Meaning	step	track	hit
<i>Identification and bookkeeping</i>					
runid	int	run number	=	=	=
evid	int	event number within the run	=	=	=

Key	Unit	Meaning	step	track	hit
trid	int	track ID	=	=	o
parent	int	parent track ID, 0 for a primary	=	=	o
det	string	sensitive detector name (GDML auxvalue)	=	=	=
detid	int	numeric detector ID	=	=	=
vol	string	physical volume at the pre-step point	=	o	o
<i>Energy, path length and light</i>					
ekin	MeV	kinetic energy at the <i>pre</i> -step point	=	o	o
ioni	MeV	ionising energy deposit, summed	=	Σ	Σ
nioni	MeV	non-ionising deposit (NIEL), summed	=	Σ	Σ
len	mm	path length, summed	=	Σ	Σ
beta	—	v/c from the pre-step energy.	=	o	o
optPhot	count	Cherenkov photons, summed	=	Σ	Σ
scintPhot	count	scintillation photons, summed	=	Σ	Σ
<i>Position</i>					
x, y, z	mm	components of the <i>post</i> -step position	=	o	o
pos	mm $\times 3$	the three above in one key	=	o	o
pre_pos	mm $\times 3$	pre-step position	=	o	o
post_pos	mm $\times 3$	post-step position (identical to pos)	=	o	o
phi, theta	rad	spherical angles of the post-step <i>position</i> vector, not of the direction	=	o	o
<i>Direction</i> (unit vectors, dimensionless)					
dx, dy, dz	—	momentum direction at the post-step point	=	*	*
post_dx, post_dy, post_dz	—	aliases of the three above	=	*	*
dir, post_dir	$\times 3$	post-step direction as one key	=	*	*
dphi, dtheta	rad	angles of the post-step direction	=	*	*
post_dphi, post_dtheta	rad	aliases of the two above	=	*	*
pre_dx, pre_dy, pre_dz	—	momentum direction at the pre-step point	=	o	o
pre_dir	$\times 3$	pre-step direction as one key	=	o	o
pre_dphi, pre_dtheta	rad	angles of the pre-step direction	=	o	o
<i>Time</i>					
pre_global_time	ns	time since the start of the event, pre-step	=	o	o
post_global_time	ns	same, post-step	=	o	o
global_time, time	ns	aliases of post_global_time	=	o	o
pre_local_time	ns	time since the track was created, pre-step	=	o	o
post_local_time	ns	same, post-step	=	o	o
local_time	ns	alias of post_local_time	=	o	o
<i>Particle identity</i> (of the stepping particle)					
part	string	particle name, e.g. e- , proton	=	=	o
pdg	int	PDG encoding	=	=	o
pdg_c	<i>e</i>	charge (Geant4 internal unit eplus)	=	=	o
pdg_m	MeV	rest mass (raw internal units)	=	=	o
part_A	int	<i>Z</i> , not <i>A</i> : GetAtomicNumber() , 0 for non-ions	=	=	o

Key	Unit	Meaning	step	track	hit
part_M	int	A: <code>GetAtomicMass()</code> , 0 for non-ions	=	=	o
<i>Primary vertex</i> (constant within an event)					
pvx, pvy, pvz	mm	primary vertex position	=	=	=
pvpos	mm ×3	the three above in one key	=	=	=
pvdxd, pvdxd, pvdz	—	primary momentum direction	=	=	=
pvdird	×3	the three above in one key	=	=	=
pvekin	MeV	primary kinetic energy	=	=	=
pvphi, pvtheta	rad	angles of the vertex <i>position</i>	=	=	=
pvdphi, pvdtheta	rad	angles of the primary <i>direction</i>	=	=	=
pvpart	string	primary particle name	=	=	=
pvpdg	int	primary PDG encoding	=	=	=
pvpart_A	int	Z of the primary, 0 for non-ions	=	=	=
pvpart_M	int	A of the primary, 0 for non-ions	=	=	=
<i>Processes</i> (name, or <code>creation</code> when null)					
pre_proc	string	process that defined the pre-step point	=	o	o
post_proc	string	process that defined the post-step point	=	*	*
proc	string	alias of <code>post_proc</code>	=	*	*
trk_proc	string	process that created the track	=	=	o

1.3 Which keys to use at which level

Step level Everything is exact. Use it for debugging, for dE/dx studies and for checking the light models — and only for short runs, since it writes one row per step.

Track level Sound for: per-particle energy deposition (`ioni`, `nioni`), path length in the detector (`len`), light contribution per particle species (`optPhot`, `scintPhot`), particle identity (`part`, `pdg`, `part_M`), creation process (`trk_proc`), entry state (`ekin`, `pre_pos`, `pre_dir`) and exit state (`post_dir`, `post_proc`). This is the level at which secondary contributions can be separated from the primary's.

Hit level Sound for: the total signal of a detector (`ioni`, `nioni`, `optPhot`, `scintPhot`, `len`) and everything about the primary (all `pv*` keys, plus `evid` and `detid`). This is the level the response functions are built from.

2 Cherenkov model — /sd/optics/*

The implementation of Cherenkov and Scintillation light is also provided from Geant4 itself. If all parameters are known and a detailed analysis is required, this is the go-to. The manual implementation with macro commands in this G4 application arose from a lack of documentation as well as convenience and sits somewhere between the rigid G4 implementation and a post-processing approach.

2.1 Formula

For a step of length Δx travelled by a particle of charge z (in units of e) and velocity β , the number of *detected* photons is the Frank–Tamm yield folded with the photocathode quantum efficiency $QE(\lambda)$:

$$N_\gamma = 2\pi\alpha z^2 \Delta x \int_{\lambda_1}^{\lambda_2} \frac{QE(\lambda)}{\lambda^2} \left[1 - \frac{1}{\beta^2 n^2(\lambda)} \right] d\lambda, \quad (1)$$

Two variants are implemented:

constant *n* (/sd/optics/const) $n(\lambda) = n$ is pulled out of the integral. Without a QE file the integral reduces to the closed form $(1/\lambda_1 - 1/\lambda_2)$, i.e. the *emitted* photon count.

dispersive (/sd/optics/disp) $n(\lambda)$ and $QE(\lambda)$ are tabulated and Eq. (1) is integrated numerically (trapezoid, step $\Delta\lambda$, default 1 nm). QE file is a required argument. For usage without, create a flat unit response, which then also defines the band

β is evaluated from the *pre-step* kinetic energy T and the PDG mass m , $\beta = \sqrt{1 - (1 + T/m)^{-2}}$.

Command	Arguments
/sd/optics/const	<det> <n> <lamMin_nm> <lamMax_nm> [qe_file]
/sd/optics/disp	<det> <refr_file> <qe_file> [dlam_nm]
/sd/optics/clear	<det> or all

2.2 What the model does not include

- No photon propagation: no reflection, absorption, wavelength shifting or light-guide geometry.
- No emission angle: the Cherenkov cone is integrated over, individual photon directions are not available.
- β is taken at the start of the step, so a strongly decelerating step slightly over-estimates the yield.

3 Scintillation model — /sd/scint/*

3.1 Formula

For each step, with dE the *ionising* energy deposit (the non-ionising part does not produce light) and dx the step length:

$$N_\gamma = \varepsilon \frac{S dE}{1 + k_B \frac{dE}{dx} + C \left(\frac{dE}{dx} \right)^2}, \quad (2)$$

Birks' law with the optional second-order Chou term. Evaluating it per step with the true dE/dx is the point of the implementation: quenching is strongly non-linear, so applying it to an event-summed energy with an assumed path length is wrong for stopping particles and for events with several tracks.

3.2 Parameters

Command	Arguments
/sd/scint/const	<det> <S_perMeV> <kB_mm/MeV> [eff]
/sd/scint/spec	<det> <S_perMeV> <kB_mm/MeV> <emission_file> <qe_file> [dlam_nm]
/sd/scint/chou	<det> <C_mm2/MeV2>
/sd/scint/clear	<det> or all

Symbol	Command argument	Meaning
S	S_perMeV	absolute light yield
k_B	kB_mm/MeV	Birks constant mm/MeV
C	/sd/scint/chou	second-order Chou term, mm ² /MeV ² (0 unset)
ε	eff	collection $\times$ quantum efficiency (1 default)
$\Delta\lambda$	dla_m_nm	grid for the spectral variant (1nm default)

Table 2: Scintillation parameters. k_B is in mm/MeV, matching Geant4’s own **SetBirksConstant** convention, so literature values apply unchanged.

In the spectral variant (**/sd/scint/spec**) ε is not a free parameter but is computed from the overlap of the scintillator emission spectrum $I(\lambda)$ with the photocathode response,

$$\varepsilon = \frac{\int I(\lambda) \text{QE}(\lambda) d\lambda}{\int I(\lambda) d\lambda}, \quad (3)$$

integrated over the emission band; light emitted outside the tabulated QE range counts in the denominator only, i.e. as emitted but undetected.

The Chou term

/sd/scint/chou sets C and nothing else. Three consequences:

- It is never standalone. Issued on its own it leaves the detector’s model unset, so that detector produces *no* light at all — silently. It must accompany a **const** or **spec** line.
- It applies to **either** model: C is stored next to the model choice, not inside it, and the quenching denominator is shared.
- The two orderings below are equivalent, because **const/spec** and **chou** write disjoint fields. The corollary is that C is sticky: re-issuing **const**, or switching to **spec**, does not reset it. Clear it with **chou <det> 0** or **/sd/scint/clear <det>**.

3.3 Caveats

- $\varepsilon = 1$ by default, i.e. **scintPhot** is the number of *emitted* photons unless an efficiency is given. Collection and quantum efficiency must be supplied either as **eff** or downstream.
- Light transport inside the scintillator (attenuation, position dependence of collection...) is not modelled.